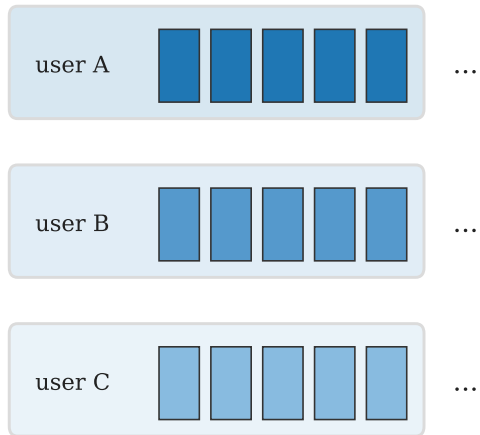


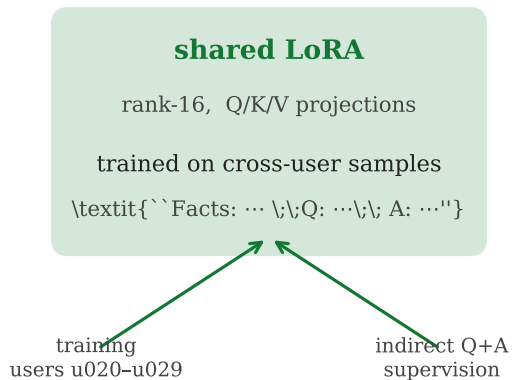
(a) Per-user content
(Engram override, 88\,KB/user)



Δ bpb on unrelated text = +0.0001

(fires only on trigger N-gram)

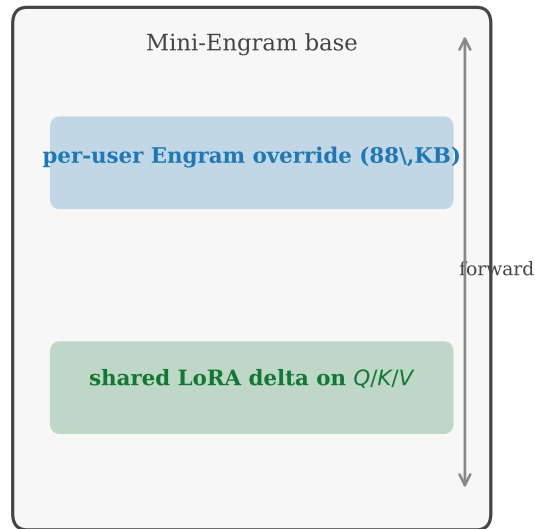
(b) Shared reasoning skill
(one LoRA, 12\,MB, amortised)



Δ bpb on unrelated text = +0.39

(amortised across 1 M+ users)

(c) At inference
(stacked, both layers live)



total per-user storage: 88\,KB